MATH 1342 – Elementary Statistics – Week 1: Introduction and the Structure of Data

Aug 24 – 26 | Introduction: What Is Statistics For? • 1.1 The Structure of Data | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe statistics as the process of collecting, describing, and analyzing data, and judge a striking result by how likely it is to occur by chance alone. (CLO 1, CLO 4) **2.** Distinguish a sample from a population and explain how a non-representative sample biases an estimate. (CLO 1) **6.** Compute and compare the mean and median of a small dataset and explain how an outlier affects each. (CLO 2)	Introduction Slides: What Is Statistics For? (Day 1)	Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age		Blackboard
3. Identify the cases and variables in a dataset presented as a table. (CLO 2) **4.** Classify a variable as categorical or quantitative. (CLO 2) **5.** Identify the explanatory and response variables in a research question. (CLO 1)	e-Text: 1.1 The Structure of Data Interactive Lecture: 1.1 The Structure of Data Slides: 1.1 The Structure of Data	Participation: Variables – identify cases, variables and variable types	Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM)	WileyPLUS StatKey Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)	START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies & Student Support Services	Participation: Syllabus – course tour and expectations		Content Area in Blackboard

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 2** Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 4** Explain the role of probability in statistics.

MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Aug 31 – Sep 2 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Distinguish a population from a sample and describe statistical inference. (CLO 1) **2.** Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CLO 1) **3.** Recognize bias from question wording, question context, and inaccurate responses. (CLO 1)	e-Text: 1.2 Sampling from a Population Interactive Lecture: 1.2 Sampling from a Population Slides: 1.2 Sampling from a Population (includes sampling video)	Participation: Variables in Statistics (Kahoot!) – Mon: review of Week 1 Participation: Sampling – spot the bias in sample scenarios Participation: Sampling Methods (Kahoot!) – Wed: sampling review	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Blackboard Kahoot!
4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CLO 1) **5.** Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CLO 1)	e-Text: 1.3 Experiments and Observational Studies Interactive Lecture: 1.3 Experiments and Observational Studies Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 1 (due Wed Sep 9, 11:59 PM) (all objectives)	Assignment Tool in Blackboard

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions.

MATH 1342 – Elementary Statistics – Week 3: Categorical Variables

Sep 7 – 9 | Labor Day – No Class (Mon Sep 7) • 2.1 Categorical Variables | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct frequency and relative frequency tables, compute a sample proportion $\hat{p}$, and distinguish the parameter p from the statistic $\hat{p}$. (CLO 2) **2.** Compute and interpret overall and conditional proportions, and a difference in proportions, from a two-way table, choosing the correct denominator. (CLO 2) **3.** Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CLO 2)	e-Text: 2.1 Categorical Variables Interactive Lecture: 2.1 Categorical Variables Slides: 2.1 Categorical Variables	Participation: Frequency and Relative Frequency Participation: Finding Proportions from Two-Way Tables	Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM)	WileyPLUS StatKey Blackboard

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data.

MATH 1342 – Elementary Statistics – Week 4: One Quantitative Variable: Shape, Center, Spread and Percentiles

Sep 14 – 16 | 2.2 One Quantitative Variable: Shape and Center • 2.3 One Quantitative Variable: Percentiles | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CLO 2) **2.** Compute the mean and median of a dataset and explain how skewness and outliers affect each (resistance). (CLO 2) **3.** Interpret the standard deviation as the typical distance from the mean and apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CLO 2)	e-Text: 2.2 One Quantitative Variable Interactive Lecture: 2.2 One Quantitative Variable Slides: 2.2 One Quantitative Variable	Participation: Quantitative Variables – shape, mean vs. median; estimate the mean and SD of a dotplot without calculating (95% rule) Participation: One Quantitative Variable (Kahoot!) – Wed: review of 2.2	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 20, 11:59 PM)	WileyPLUS StatKey Blackboard Kahoot!
4. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **5.** Compute and interpret z-scores to compare values measured on different scales or in different groups. (CLO 2) **6.** Find the five-number summary, range, and IQR; apply the 1.5×IQR rule to identify outliers; and construct or interpret a boxplot. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: Using Percentiles – Your Turn: comparing heights with z-scores; five-number summary	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 2 (due Mon Sep 21, 11:59 PM) (all objectives)	Assignment Tool in Blackboard

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data.

MATH 1342 – Elementary Statistics – Week 5: Two-Variable Relationships and Linear Regression

Sep 21 – 23 | 2.4 Two Variable Relationships • 2.5 Two Quantitative Variables: Linear Regression | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CLO 2) **2.** Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CLO 7) **3.** Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Participation: Box Plots and Outliers – Mon: 1.5×IQR rule review; side-by-side boxplots (tea vs. coffee)	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 27, 11:59 PM)	WileyPLUS Blackboard
4. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CLO 7) **5.** Use a regression equation to predict a response value, and compute and interpret a residual. (CLO 7) **6.** Use technology to find the least-squares regression line and explain what “least squares” means. (CLO 7) **7.** Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression	Examples: baseball game length and commute-time regression lines (predict, interpret slope, find a residual)	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)		Participation: Two Variable Relationships – Wed: NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier; Your Turn: spotting residuals (MO2, MO3, MO5)		
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 2 Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 7** Solve linear regression and correlation problems.

MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 28 – 30 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. List the sample space and count outcomes to compute probabilities for equally likely outcomes, and apply the complement rule. (CLO 3) **2.** Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)	WileyPLUS StatKey Blackboard
3. Classify a numerical summary as a parameter or a statistic and use the correct notation (μ, σ, p, ρ vs. $\bar{x}$, s, $\hat{p}$, r). (CLO 5) **4.** Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CLO 5) **5.** Explain the standard error as the variability of a statistic and predict how sample size affects it. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents	Online Homework: 3.1 Sampling Distributions (due Sun Oct 11, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 3 (due Mon Oct 5, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)		Participation: Regression Line – Mon: used-car price problem wrapping up 2.5 (Week 5 objectives)	Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables.

MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 5 – 7 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct a 95% confidence interval for a mean or proportion as statistic ± 2·SE, given the statistic and its standard error. (CLO 6) **2.** Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CLO 4, CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS Blackboard
3. Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CLO 5, CLO 6) **4.** Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CLO 5, CLO 6) **5.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator) StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	Participation: Bootstrap Confidence Intervals – Wed: Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)	WileyPLUS Blackboard StatKey
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 4 (due Mon Oct 12, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)		Participation: Sampling Distribution (Kahoot!) – Mon: review of 3.1 (Week 6 objectives)	Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)	Kahoot! Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 4 Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.

MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests

Oct 12 – 14 | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CLO 3, CLO 6) **2.** Predict how changing the confidence level or the sample size affects the width of an interval. (CLO 6) **3.** Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CLO 6)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	Participation: Confidence Intervals Using Percentiles – Mon: StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM)	WileyPLUS StatKey Blackboard
4. Identify the parameter of interest and write null and alternative hypotheses from a research question. (CLO 8) **5.** Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming H_0 and asking whether the data are surprising. (CLO 8)	e-Text: 4.1 Introducing Hypothesis Tests Interactive Lecture: 4.1 Introducing Hypothesis Tests Slides: 4.1 Introducing Hypothesis Tests	Participation: Writing Hypotheses – Rock paper scissors practice (9 of 11 wins): define the parameter, state H_0 and H_a, choose the tail Think about it: is 15 of 16 correct (dolphin study) convincing?	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM)	WileyPLUS StatKey Blackboard

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 6** Describe and compute confidence intervals.
CLO 8 Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance

Oct 19 – 21 | 4.2 Measuring Evidence with P-Values • 4.3 Determining Statistical Significance | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a randomization distribution is generated assuming H_0 is true, and identify its center as the null value. (CLO 5, CLO 8) **2.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) **3.** Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from H_a. (CLO 3, CLO 8) **4.** Interpret a p-value in context and relate its size to the strength of evidence against H_0. (CLO 4, CLO 8) **5.** Compare the p-value to α, decide whether to reject H_0, write a two-step conclusion in context, and avoid saying "accept H_0". (CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes "What is a p-value?" video) StatKey – Randomization Hypothesis Tests	Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed Participation: P-values and Conclusions – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM) Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Mon Oct 26, 11:59 PM)	WileyPLUS StatKey Blackboard Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Practice Test 2 (Blackboard, due Thu Oct 29, 11:59 PM; 120 min, proctoring practice)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing, Making Connections, and Normal Distributions

Oct 26 – 28 | 4.4 A Closer Look at Testing • 4.5 Making Connections • 5.1 Hypothesis Testing Using Normal Distributions | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CLO 4, CLO 8) **2.** Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM) Online Homework: 4.4 A Closer Look at Testing (b) (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping), and compare bootstrap and randomization distributions by their center and use. (CLO 5, CLO 8) **4.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05, and decide whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
5. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution N(null value, SE). (CLO 5) **6.** Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 6 (due Mon Nov 2, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 11: Inference for One Proportion and Two Proportions

Nov 2 – 4 | 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) • 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **2.** Construct and interpret a confidence interval for a population proportion with z* for 90%, 95%, or 99% confidence. (CLO 6) **3.** Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 6.1-CI Confidence Interval for a Proportion e-Text: 6.1-D Distribution of a Proportion e-Text: 6.1-HT Hypothesis Test for a Proportion Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	WileyPLUS StatKey Blackboard
4. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CLO 5) **5.** Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **6.** Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3-CI Confidence Interval for a Difference in Proportions e-Text: 6.3-D Distribution of a Difference in Proportions e-Text: 6.3-HT Hypothesis Test for a Difference in Proportions Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 8, 11:59 PM)	WileyPLUS StatKey Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 5 Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 12: Inference for a Mean

Nov 9 – 11 | 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) • Veterans Day – No Class (Wed Nov 11) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain why inference for a mean uses the t distribution with df = n − 1 instead of the normal distribution. (CLO 5) **2.** Check the conditions for t procedures (n ≥ 30, or roughly normal data without extreme outliers). (CLO 5) **3.** Construct and interpret a t confidence interval for a population mean, including the margin of error. (CLO 6) **4.** Carry out a one-sample t test for μ and state the conclusion in context. (CLO 8)	e-Text: 6.2-CI Confidence Interval for a Mean e-Text: 6.2-D Distribution of a Mean e-Text: 6.2-HT Hypothesis Test for a Mean Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM) Take-Home Quiz 7 (due Mon Nov 16, 11:59 PM)	WileyPLUS StatKey Blackboard Assignment Tool in Blackboard

CLO 5 Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 13: Inference for Two Means and Paired Data

Nov 16 – 18 | 6.4 Inference for a Difference in Means (Distribution, CI, Hypothesis Test) • 6.5 Paired Difference in Means | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the standard error and conservative degrees of freedom for a difference in two independent means, and check the conditions. (CLO 5) **2.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$, explaining how the order of the groups affects the sign. (CLO 6) **3.** Carry out a two-sample t test, choosing the tail from the research question. (CLO 8)	e-Text: 6.4-CI Confidence Interval for a Difference in Means e-Text: 6.4-D Distribution of a Difference in Means e-Text: 6.4-HT Hypothesis Test for a Difference in Means Interactive Lecture: 6.4 Inference for a Difference in Means Slides: 6.4 Inference for a Difference in Means	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)	WileyPLUS StatKey Blackboard
4. Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8) **5.** Define the difference variable d, carry out a paired t test and paired t confidence interval, and explain why pairing reduces variability. (CLO 6, CLO 8)	e-Text: 6.5 Paired Difference in Means Interactive Lecture: 6.5 Paired Difference in Means Slides: 6.5 Paired Difference in Means	Participation: Paired or Independent? – class discussion of eight study scenarios (a)–(h); define d and the order of subtraction	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Test 3 – Fri Nov 20 – Sun Nov 22 (Blackboard, proctored, 2 hours). Covers Weeks 10–13. (all objectives) Take-Home Quiz 8 (due Mon Nov 23, 11:59 PM) (all objectives)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor Assignment Tool in Blackboard

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 14: Chi-Square Test for Association

Nov 23 – 25 | 7.2 Testing for an Association Between Two Categorical Variables • No Class – College closes at noon (Wed Nov 25) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. State the hypotheses for a chi-square test of association from a two-way table. (CLO 8) **2.** Compute expected counts and check that all expected counts are at least 5. (CLO 8) **3.** Compute the chi-square statistic and df = (r − 1)(c − 1), and interpret the right-tail p-value. (CLO 8) **4.** Relate the chi-square test on a 2×2 table to the two-proportion z test, and explain that a significant result shows association but not direction or causation. (CLO 8)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables Slides: 7.2 Testing for an Association Between Two Categorical Variables StatKey – Theoretical Distributions (χ^2)	Participation: Chi-Square Test – faculty rank × gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	WileyPLUS StatKey Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 8 Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 15: ANOVA and Inference for Slope and Correlation

Nov 30 – Dec 2 | 8.1 Analysis of Variance • 9.1 Inference for Slope and Correlation | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Identify when ANOVA is appropriate and state H_0 and H_a in symbols and words. (CLO 8) **2.** Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against H_0. (CLO 8) **3.** Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CLO 8)	e-Text: 8.1 Analysis of Variance Interactive Lecture: 8.1 Analysis of Variance Slides: 8.1 Analysis of Variance StatKey – Theoretical Distributions (F)	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard
4. State the simple linear model, distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r, and interpret the slope in context. (CLO 7) **5.** Use regression output to test H_0: $\beta_1 = 0$ (equivalently $\rho = 0$) with a t statistic on $n - 2$ degrees of freedom, state the conclusion in context, and explain why the slope and correlation tests give the same p-value. (CLO 7, CLO 8) **6.** Construct and interpret a confidence interval for the slope, interpret R^2 as the proportion of variability explained, and check the conditions with a residual plot. (CLO 6, CLO 7)	e-Text: 9.1 Inference for Slope and Correlation Interactive Lecture: 9.1 Inference for Slope and Correlation Slides: 9.1 Inference for Slope and Correlation	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); self-quiz: drinks per week vs. classes missed	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Course Evaluation (in START HERE) – complete before the end of the semester	

CLO 6 Describe and compute confidence intervals. **CLO 7** Solve linear regression and correlation problems. **CLO 8** Perform hypothesis testing using statistical methods.

MATH 1342 – Elementary Statistics – Week 16: Putting It All Together and Final Exam

Dec 7 – 9 | Review: Putting It All Together – From Sampling to Hypothesis Testing (Mon Dec 7) • Final Exam – Wed Dec 9, online (Blackboard, proctored) | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Choose the correct procedure among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression, and carry out a complete six-step hypothesis test. (CLO 8) **2.** Carry out and interpret a complete confidence interval or hypothesis test, using simulation or theoretical distributions as appropriate. (CLO 6, CLO 8) **3.** Interpret results in context, including what the study design allows the conclusion to claim about generalization and causation. (CLO 1)	Review Slides: Putting It All Together (with answer key and no-answers versions) All Slides and Interactive Lectures (Weeks 1–15) StatKey	Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI	Final Exam – Wed Dec 9 (Blackboard, proctored, 2 hours, cumulative; no retake)	Blackboard StatKey Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.